7.6 Inverse Trigonometric Functions

The six basic trigonometric functions are not 1-1. So to have sensible meaning for their inverses, we can restrict their domains to intervals on which they are 1-1.

For example, by restricting the domain to $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the sine function is 1-1, and hence has an inverse.

Under suitable restricted domains, these trigonometric functions are 1-1, and they have inverses, which we can denote as follows.

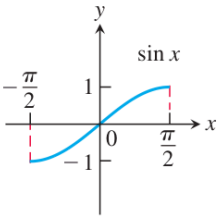
$$y = \sin^{-1} x \text{ or } y = \arcsin x \qquad y = \cos^{-1} x \text{ or } y = \arccos x$$

$$y = \tan^{-1} x \text{ or } y = \arctan x \qquad y = \cot^{-1} x \text{ or } y = \operatorname{arccot} x$$

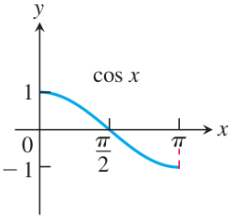
$$y = \sec^{-1} x \text{ or } y = \operatorname{arcsec} x \qquad y = \csc^{-1} x \text{ or } y = \operatorname{arccsc} x$$

The graphs of trigonometric functions under suitable restricted domains.

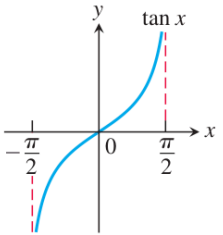
Domain restrictions that make the trigonometric functions one-to-one



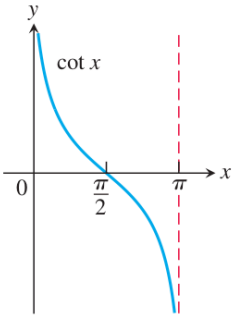
$y = \sin x$
Domain: $[-\pi/2, \pi/2]$
Range: $[-1, 1]$



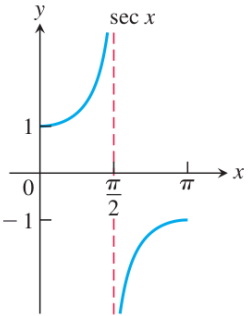
$y = \cos x$
Domain: $[0, \pi]$
Range: $[-1, 1]$



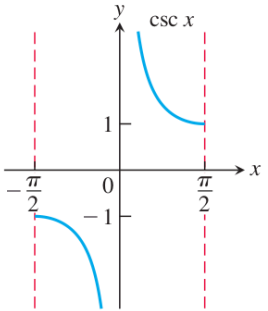
$y = \tan x$
Domain: $(-\pi/2, \pi/2)$
Range: $(-\infty, \infty)$



$y = \cot x$
Domain: $(0, \pi)$
Range: $(-\infty, \infty)$

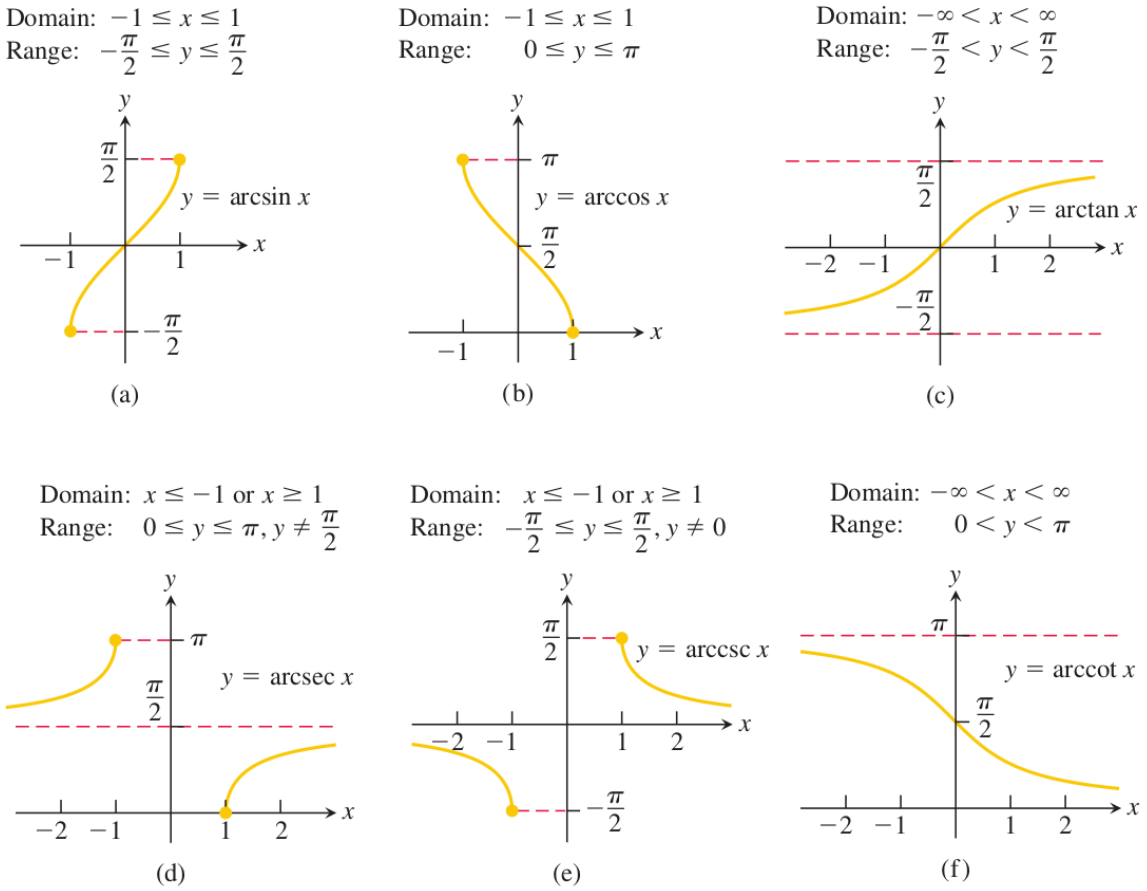


$y = \sec x$
Domain: $[0, \pi/2) \cup (\pi/2, \pi]$
Range: $(-\infty, -1] \cup [1, \infty)$



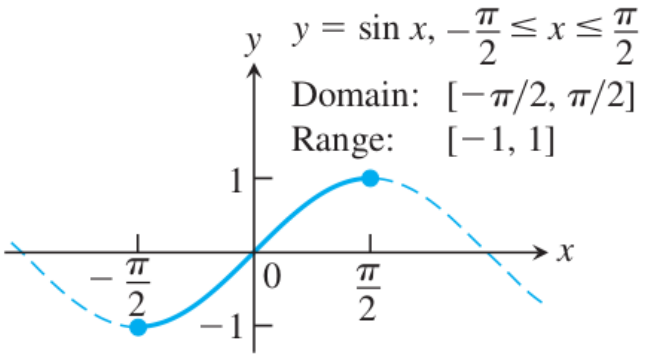
$y = \csc x$
Domain: $[-\pi/2, 0) \cup (0, \pi/2]$
Range: $(-\infty, -1] \cup [1, \infty)$

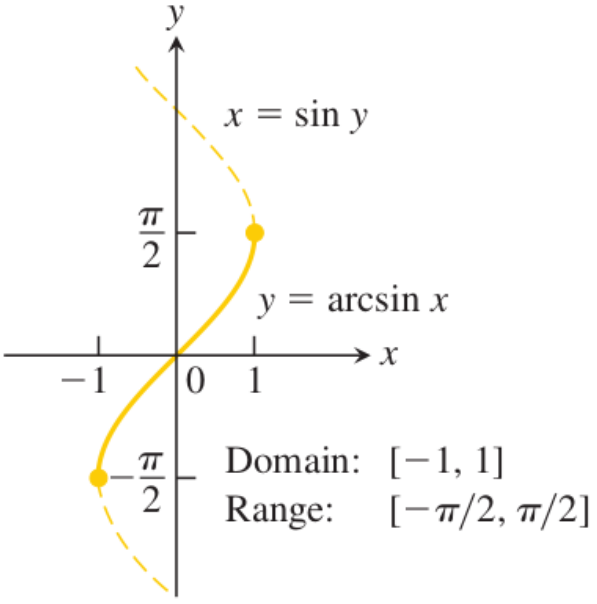
The graphs of inverse trigonometric functions.



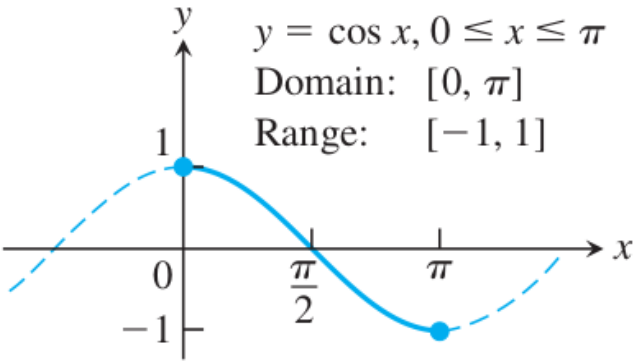
The Arcsine and Arccosine Functions

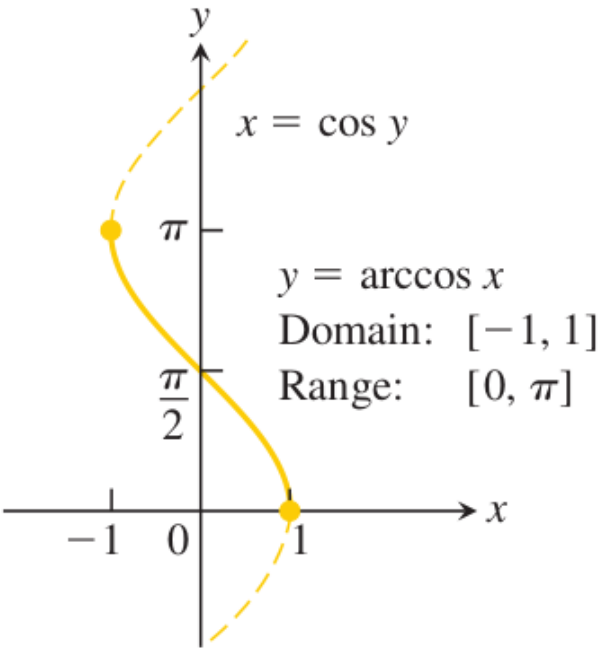
The function sin x is one-to-one on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so the inverse of sine exists. We let $y = \sin^{-1}(x)$ to be the value in $[-\frac{\pi}{2}, \frac{\pi}{2}]$ for which $\sin y = x$.





The function $\cos x$ is one-to-one on $[0, \pi]$, so the inverse of cosine exists. We let $y = \cos^{-1}(x)$ to be the value in $[0, \pi]$ for which $\cos y = x$.





Identities involving Arcsine and Arccosine

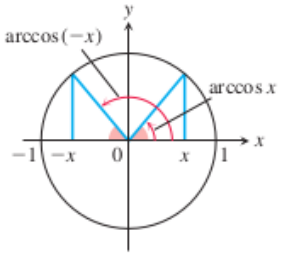


FIGURE 7.27 arccos x and arccos $(-x)$ are supplementary angles (so their sum is π).

$$\sin^{-1}(-x) = -\sin^{-1}(x)$$

$$\cos^{-1}(x) + \cos^{-1}(-x) = \pi$$

$$\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$$

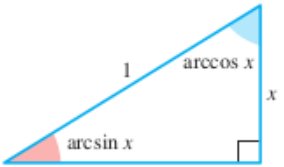
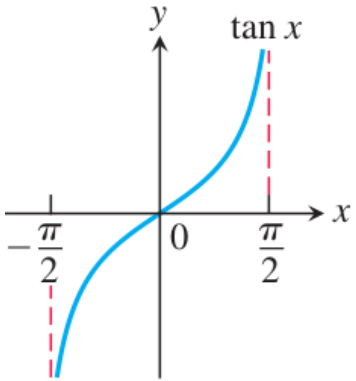
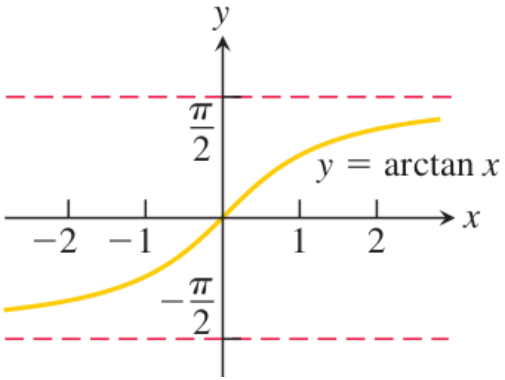


FIGURE 7.28 arcsin x and arccos x are complementary angles (so their sum is $\pi/2$).

The Arctangent and Arccotangent Functions

The function $\tan x$ is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$, so the inverse of tangent exists.





The function $\cot x$ is one-to-one on $(0, \pi)$, so the inverse of cotangent exists.

Identities involving Arctangent and Arccotangent

$$\tan^{-1}(-x) = -\tan^{-1}x$$ (Arctangent is odd)

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

Example:

a. Find sec $(\tan^{-1}\frac{x}{3})$.

b. Find sin $(\tan^{-1}\sqrt{x^2 - 2x})$.

The inverses of secant and cosecant functions

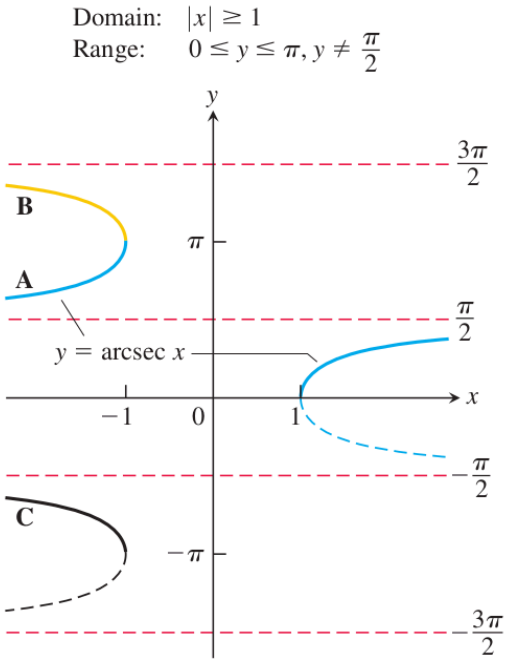


FIGURE 7.29 There are several logical choices for the left-hand branch of $y = \text{arcsec } x$. With choice **A**, $\text{arcsec } x = \arccos(1/x)$, a useful identity employed by many calculators.

By restricting the domain to $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$, the function sec x is 1-1. So $\sec^{-1}(x)$ exists for $|x| \geq 1$. Under such restrictions, we have the following identity.

$$\sec^{-1}(x) = \cos^{-1}(\frac{1}{x}) = \frac{\pi}{2} - \sin^{-1}(\frac{1}{x})$$

By restricting the domain to $[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$, the function csc x is 1-1. So $\csc^{-1}(x)$ exists, for $|x| \geq 1$.

Identities Involving Inverse Trigonometric:

1. $\sin^{-1}(-x) = -\sin^{-1}x$	2. $\tan^{-1}(-x) = -\tan^{-1}x$
3. $\cos^{-1}x + \cos^{-1}(-x) = \pi$	4. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$
5. $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$	6. $\sec^{-1}x + \csc^{-1}x = \frac{\pi}{2}$
7. $\sec^{-1}x = \cos^{-1}\frac{1}{x}$	8. $\csc^{-1}x = \sin^{-1}\frac{1}{x}$
9. $\cot^{-1}x = \tan^{-1}\frac{1}{x}$	

The Derivative of $y = \sin^{-1} u$

Let $f(x) = \sin x$, then $f^{-1}(x) = \sin^{-1} x$.

Example:

1. $\frac{d}{dx}(\sin^{-1}x^2) =$

2. $\frac{d}{dx}(\sin^{-1}\sqrt{1-x}) =$

The Derivative of $y = \tan^{-1} u$

Let $f(x) = \tan x$, then $f^{-1}(x) = \tan^{-1} x$.

Example:

1. $\frac{d}{dx}\tan^{-1}(1 + x^2) =$

2. $\frac{d}{dx}\ln(\tan^{-1}x) =$

The Derivative of $y = \sec^{-1} u$

Instead of applying the inverse derivative theorem as before, we find the derivative of $y = \sec^{-1} x,\ |x| > 1$, using implicit differentiation and the Chain Rule:

$$y = \sec^{-1} x$$

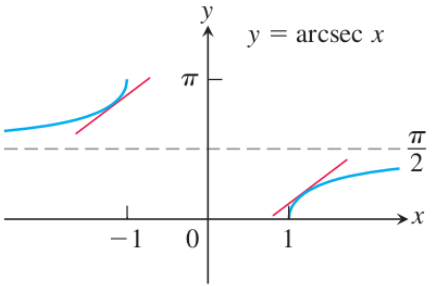


FIGURE 7.31 The slope of the curve $y = \text{arcsec}\, x$ is positive for both $x < -1$ and $x > 1$.

The Derivative of $y = \sec^{-1} u$

We can also get the derivative of $\sec^{-1}(x)$ using the identity $\sec^{-1}(x) = \cos^{-1}(\frac{1}{x}) = \frac{\pi}{2} - \sin^{-1}(\frac{1}{x})$.

Derivative of the Other Three

The derivatives of the other three trigonometric functions are easier to derive using the following identities.

Inverse Function-Inverse Cofunction identities

$$\cos^{-1}x = \frac{\pi}{2} - \sin^{-1}x$$

$$\cot^{-1}x = \frac{\pi}{2} - \tan^{-1}x$$

$$\csc^{-1}x = \frac{\pi}{2} - \sec^{-1}x$$

TABLE 7.3 Derivatives of the inverse trigonometric functions

1. $\dfrac{d(\arcsin u)}{dx} = \dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$

2. $\dfrac{d(\arccos u)}{dx} = -\dfrac{1}{\sqrt{1-u^2}}\dfrac{du}{dx}, \quad |u| < 1$

3. $\dfrac{d(\arctan u)}{dx} = \dfrac{1}{1+u^2}\dfrac{du}{dx}$

4. $\dfrac{d(\operatorname{arccot} u)}{dx} = -\dfrac{1}{1+u^2}\dfrac{du}{dx}$

5. $\dfrac{d(\operatorname{arcsec} u)}{dx} = \dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$

6. $\dfrac{d(\operatorname{arccsc} u)}{dx} = -\dfrac{1}{|u|\sqrt{u^2-1}}\dfrac{du}{dx}, \quad |u| > 1$

Example: Find

1. $\frac{d}{dx}\sin^{-1}\sqrt{1-x},\ 0 < x < 1.$

2. $\frac{d}{dx}\tan^{-1}(1+x^2)$

3. $\frac{d}{dx}\sec^{-1}(\frac{e^{2x}}{2}),\ x > \frac{1}{2}.$

4. $\frac{d}{dx}\sec^{-1}(\frac{1}{x}),\ 0 < x < 1.$

5. $\frac{d}{dx}\csc^{-1}(x^3)$

6. $\frac{d}{dx}\cos^{-1}(\frac{x-1}{x+1})$

Integrals evaluated with inverse trigonometric functions

TABLE 7.4 Integrals evaluated with inverse trigonometric functions

The following formulas hold for any constant $a > 0$.

1. $\displaystyle\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$ (Valid for $u^2 < a^2$)

2. $\displaystyle\int \frac{du}{a^2 + u^2} = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right) + C$ (Valid for all u)

3. $\displaystyle\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{u}{a}\right| + C$ (Valid for $|u| > a > 0$)

Example:

1. $\int \frac{dx}{\sqrt{3-4x^2}} =$

2. $\int \frac{dx}{\sqrt{4x-x^2}} =$

3. $\int \frac{dx}{4x^2+4x+2} =$

4. $\int \frac{dx}{\sqrt{e^{2x}-6}} =$